actual data reporting of each machine, idle or moving.

However, the company faced several challenges to integrate their product with different machines (be it construction equipment, power back utilities, trucking and big storage tanks).

Getting the right customers and the right price for the technology are some other limitations. The company is looking for strategic investors and board advisors. So far, FuelArk has created a niche in the infrastructure and construction space. Some leading brands in India are already using their offerings.

Real-life application

Following benefits were achieved by a manufacturing company by implementing FuelArk:

- Replacement of manual fuel consumption log books by automated reconciliation processes.
- Different gensets showed different consumption levels of fuel while functioning at exactly the same load. This resulted in identification of gensets that showed sub-optimal performance and needed maintenance servicing. It finally resulted in lesser consumption of fuel after proper servicing of those gensets.

Solar Streetlights With Inbuilt Lithium-Ion Batteries To The Rescue

Intelizon was founded by Dr Kushant Uppal, a global professional with 20-plus years of techno-commercial success. He returned to India to develop a unique solar-based light system to light up our streets. Wireless outdoor lighting solutions are at the core of Intelizon's success and innovation.

Zonstreet by Intelizon, with an integrated lithium-ion battery and an inbuilt sensor, turns on in the evening and provides 10 to 12 hours of lighting at night

Solar streetlight is a US$ 3 to US$ 4 billion market globally. It is estimated to grow to US$ 15 billion by 2024. The market is dominated by regional and local players with customised specifications.

The market has faced many field issues over the years. Companies like Intelizon are proving the reliability that has led to the acceptance of lithium-ion batteries. (Recently, MNRE mandated all state government to stop using lead-acid batteries and transition to lithium-ion ones for solar streetlights). Quite a few brands are expected to emerge in the next three to five years.

Lack of reliable products lead to the focus on lithium-ion-based solar systems. In the case of streetlights, lead-acid solar systems had theft as well as performance issues, leading to field failures.

Intelizon already had innovative and reliable home lighting systems, but their channel partners asked for inbuilt lithium-ion-based solar streetlights to address the field issues. Intelizon, with their Gambia partner, initiated the idea for inbuilt lithium-ion-based solar streetlights.

The process

Designing a reliable system was the main challenge for Intelizon team. They faced installation-related as well as environmental issues. These were addressed with redesign. Today, they have systems working in the field for more than five years.

Solar panels charge the battery during the day. The battery then provides power to the LED, which runs all night. Intelizon has auto-dimming, self-autonomy (efficient charging and performance even on cloudy days), remote monitoring and control as well as inbuilt CCTVs as some of the key features in their models. There is a strong technology element with optics (LED and lenses), solar, battery, firmware, hardware and software coming together to create advanced and reliable systems.

Operational limitations. The company currently limits the operating conditions between 0°C and 50°C. There are certain regions globally that require extreme temperature operations at sub-zero or 60°C.

Incubation and funding. Intelizon was incubated at IIT-Madras, and raised venture capital to develop the technology for home lighting systems. Initial product knowledge was leveraged to develop more complex outdoor lighting systems. The company did not raise further funds and took about one year to design the system and another year to make it field reliable.

Consumer benefits

Outdoor solar lights have a great value proposition in Greenfield locations where underground cabling costs can be saved. There is also additional savings on electricity and maintenance costs, and protection from power outages. Some applications where Intelizon lights have been installed are urban communities, commercial offices, schools, colleges, hospital campuses, rural areas, farmhouses, streets and factories. The target customers are private companies as well as government agencies.

Marketing the solution

The company appoints local channel partners globally who work with